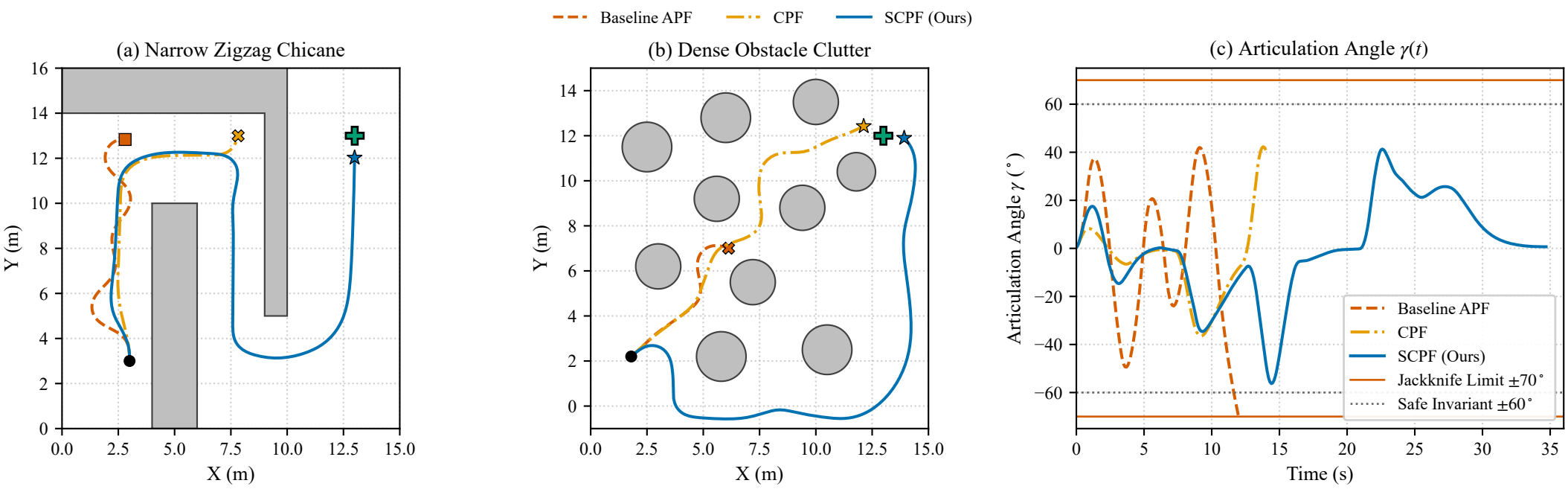

Baseline APF
CPF
SCPF (Ours)
(a) Narrow Zigzag Chicane
(b) Dense Obstacle Clutter
(c) Articulation Angle $\gamma(t)$
X (m)
Y (m)
Time (s)
Articulation Angle γ (°)
Jackknife Limit ±70°
Safe Invariant ±60°